Engineering Data

APPENDIX B

This section includes a number of tables and nomographs for conversion to units other than the SI units generally used in this paper. A summary of SI units, prefixes and symbols is also presented.

Tables of dimensions of steel pipe to ANSI, BS and ISO standards are provided as well as complete solutions of water and air flow pressure drop problems.

INTERNATIONAL SYSTEM OF UNITS (SI)

The name Système International d'Unités (International System of Units), with abbreviation SI, was adopted by the 11th General Conference of Weights and Measures in 1960.

This system includes three classes of units:

(1) base units
(2) supplementary units
(3) derived units

Together these form the coherent system of SI units.

BASE UNITS

Quantity	Name	Symbol
length	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temperature	kelvin	K
luminous intensity	candela	cd
amount of substance	mole	mol

SUPPLEMENTARY UNITS

Quantity	Name	Symbol
Plain Angle	radian	rad
Solid Angle	steradian	sr

DERIVED UNITS

Quantity	Name	Symbol	Equivalents
frequency	hertz	Hz	1 Hz = 1 cycle/s
force	newton	N	1 N = 1 kg/m/s^2
pressure and stress	pascal	Pa	1 Pa = 1 N/m^2
work, energy, quantity of heat	joule	J	1 J = 1 N m
power	watt	W	1 W = 1 J/s
quantity of electricity	coulomb	C	1 C = 1 A s
electric potential, potential difference, tension, electromotive force	volt	V	1 V = 1 W/A
electric capacitance	farad	F	1 F = 1 A s/v
electric resistance	ohm	Ω	1 Ω = 1 V/A
electric conductance	siemens	S	1 S = 1 Ω^{-1}
flux of magnetic induction, magnetic flux	weber	Wb	1 Wb = 1 V s
magnetic flux density, magnetic induction	tesla	T	1 T = 1 Wb/m^2
inductance	henry	H	1 H = 1 V s/A
luminous flux	lumen	lm	1 lm = 1 cd sr
illumination	lux	lx	1 lx = 1 lm/m^2

INTERNATIONAL SYSTEM OF UNITS (SI) (Cont'd)

EXCEPTIONS

Certain units which are outside the SI system but have international recognition and use, will continue to be used. The most important of these are:

TIME: In addition to the second (s) the following units will also continue in use:

Name	Symbol
minute	min
hour	h
day	d

Other units such as week, month and year will also continue in use.

PLANE ANGLE:

In addition to the radian (rad) the following units will continue to be used:

Name	Symbol
degree	°
minute	′
second	″

TEMPERATURE:

In addition to the kelvin (K), which relates to the absolute or thermodynamic scale, customary temperatures will be measured in degrees Celsius (° C), formerly called centigrade. The degree intervals on the Kelvin and Celsius scales are identical, but, whereas 0 Kelvin is absolute zero, 0 degrees Celsius is the temperature of melting ice.

DECIMAL MULTIPLES AND SUB-MULTIPLES OF SI UNITS – PREFIXES:

Factor		
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10	deca	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

When a prefix is added to a unit it's considered to be combined with that unit, forming a new unit symbol which can be raised to a positive or negative power and which can be combined with other unit symbols to form compound units. When a combined prefix and symbol is raised to a positive (or negative) power they must be considered as one whole individual unit and not as separate entities.

WRITTEN USE OF SYMBOLS AND PREFIXES

Primary units are spaced apart,

e.g. N m (newton meter)
kW h (kilowatt hour)

Prefixes are placed immediately adjacent to the unit.

e.g. MN (meganewton)
kJ (kilojoule)

Flow of Water Through Schedule 40 Steel Pipe

Pressure Drop per 100 metres and Velocity in Schedule 40 Pipe for Water at 15 C

Discharge	1/8"		1/4"		3/8"		1/2"		3/4"		1"		1 1/4"		1 1/2"	
Liters per Minute	Velocity Meters per Second	Press. Drop bars	Velocity Meters per Second	Press. Drop bars	Velocity Meters per Second	Press. Drop bars	Velocity Meters per Second	Press. Drop bars	Velocity Meters per Second	Press. Drop bars	Velocity Meters per Second	Press. Drop bars	Velocity Meters per Second	Press. Drop bars	Velocity Meters per Second	Press. Drop bars
1	0.459	0.726	0.251	0.17												
2	0.918	2.59	0.501	0.60	0.272	0.136	0.170	0.044								
3	1.38	5.59	0.752	1.22	0.407	0.29	0.255	0.091	0.144	0.023						
4	1.84	9.57	1.00	2.09	0.543	0.48	0.340	0.151	0.192	0.038	0.120	0.012				
5	2.29	14.45	1.25	3.18	0.679	0.70	0.425	0.223	0.241	0.057	0.150	0.017				
6	2.75	20.29	1.50	4.46	0.815	0.98	0.510	0.309	0.289	0.077	0.180	0.024				
8	3.67	35.16	2.01	7.36	1.09	1.69	0.680	0.524	0.385	0.129	0.240	0.041	0.138	0.011		
10			2.51	11.81	1.36	2.52	0.850	0.798	0.481	0.193	0.300	0.061	0.172	0.015	0.127	0.008
15			3.76	25.67	2.04	5.37	1.28	1.69	0.722	0.403	0.450	0.124	0.258	0.032	0.190	0.015
20	2"		2 1/2"		2.72	9.24	1.70	2.84	0.962	0.683	0.600	0.210	0.344	0.054	0.254	0.026
30	0.231	0.016					2.55	6.17	1.44	1.45	0.900	0.442	0.517	0.114	0.380	0.053
40	0.308	0.027	0.216	0.010			3.40	10.72	1.92	2.50	1.20	0.758	0.689	0.193	0.507	0.091
50	0.385	0.039	0.270	0.017					2.41	3.83	1.50	1.14	0.861	0.290	0.634	0.135
60	0.462	0.055	0.324	0.023					2.89	5.41	1.80	1.61	1.03	0.400	0.761	0.187
70	0.539	0.098	0.378	0.031	3"		3 1/2"		3.37	7.27	2.10	2.15	1.21	0.541	0.888	0.248
80	0.616	0.092	0.432	0.039	0.280	0.014			3.85	9.27	2.40	2.76	1.38	0.690	1.01	0.315
90	0.693	0.115	0.486	0.048	0.315	0.017	0.235	0.008			2.70	3.47	1.55	0.862	1.14	0.397
100	0.770	0.141	0.540	0.059	0.350	0.020	0.261	0.010	4"		3.00	4.25	1.72	1.05	1.27	0.488
150	1.15	0.295	0.810	0.125	0.524	0.042	0.392	0.021	0.304	0.011	4.50	9.30	2.58	2.26	1.90	1.03
200	1.54	0.512	1.08	0.212	0.699	0.072	0.523	0.036	0.405	0.019	5"		3.44	3.91	2.54	1.81
250	1.92	0.773	1.35	0.322	0.874	0.108	0.653	0.053	0.507	0.028					3.17	2.74
300	2.31	1.10	1.62	0.449	1.05	0.152	0.784	0.074	0.608	0.040	0.387	0.014	6"		3.80	3.82
350	2.69	1.47	1.89	0.606	1.22	0.203	0.915	0.099	0.710	0.053	0.452	0.018			4.44	5.18
400	3.08	1.92	2.16	0.780	1.40	0.264	1.05	0.128	0.811	0.068	0.516	0.023	0.357	0.009	5.07	6.69
450	3.46	2.39	2.43	0.979	1.57	0.329	1.18	0.161	0.912	0.084	0.581	0.028	0.402	0.012	5.71	8.45
500	3.85	2.95	2.70	1.20	1.75	0.403	1.31	0.196	1.01	0.101	0.646	0.034	0.447	0.014		
550	4.23	3.55	2.97	1.44	1.92	0.479	1.44	0.232	1.11	0.122	0.710	0.041	0.491	0.016		
600	4.62	4.20	3.24	1.69	2.10	0.566	1.57	0.273	1.22	0.146	0.775	0.047	0.536	0.019		
650	5.00	6.88	3.51	1.97	2.27	0.658	1.70	0.319	1.32	0.169	0.839	0.055	0.581	0.022		
700	5.39	5.63	3.78	2.28	2.45	0.759	1.83	0.368	1.42	0.194	0.904	0.063	0.625	0.025		
750	5.77	6.44	4.05	2.60	2.62	0.863	1.96	0.420	1.52	0.218	0.968	0.072	0.670	0.029	8"	
800			4.32	2.95	2.80	0.977	2.09	0.473	1.62	0.246	1.03	0.081	0.715	0.032		
850			4.59	3.31	2.97	1.09	2.22	0.528	1.72	0.277	1.10	0.091	0.760	0.036	0.439	0.009
900					3.15	1.22	2.35	0.585	1.82	0.308	1.16	0.010	0.804	0.041	0.465	0.010
950					3.32	1.35	2.48	0.649	1.93	0.342	1.23	0.111	0.849	0.045	0.491	0.012
1000					3.5	1.50	2.61	0.714	2.03	0.377	1.29	0.122	0.894	0.049	0.516	0.013
1100					3.85	1.75	2.87	0.860	2.23	0.452	1.42	0.147	0.983	0.059	0.568	0.015
1200					4.20	2.14	3.14	1.02	2.43	0.534	1.55	0.172	1.07	0.069	0.620	0.018
1300							3.40	1.19	2.64	0.627	1.68	0.200	1.16	0.080	0.671	0.021
1400							3.66	1.37	2.84	0.722	1.81	0.232	1.25	0.091	0.723	0.024
1500							3.92	1.56	3.04	0.818	1.94	0.264	1.34	0.105	0.775	0.027
1600	10"						4.18	1.78	3.24	0.924	2.07	0.297	1.43	0.118	0.826	0.031
1700							4.44	1.99	3.45	1.04	2.19	0.331	1.52	0.132	0.878	0.035
1800	0.590	0.012							3.65	1.16	2.32	0.369	1.61	0.147	0.930	0.039
1900	0.622	0.014							3.85	1.28	2.45	0.410	1.70	0.163	0.981	0.042
2000	0.655	0.015							4.05	1.41	2.58	0.452	1.79	0.181	1.03	0.046
2200	0.721	0.018	12"						4.46	1.70	2.84	0.545	1.97	0.217	1.14	0.056
2400	0.786	0.021									3.10	0.645	2.14	0.253	1.24	0.065
2600	0.852	0.025	0.600	0.010							3.36	0.749	2.32	0.296	1.34	0.076
2800	0.917	0.028	0.646	0.012	14"						3.61	0.859	2.50	0.339	1.45	0.087
3000	0.983	0.032	0.692	0.013	0.573	0.008					3.87	0.982	2.68	0.387	1.55	0.099
3500	1.15	0.043	0.810	0.018	0.668	0.011					4.52	1.33	3.13	0.526	1.81	0.134
4000	1.31	0.055	0.923	0.023	0.764	0.014	16"				5.16	1.72	3.57	0.673	2.07	0.172
4500	1.47	0.068	1.04	0.029	0.860	0.018	0.658	0.009					4.02	0.853	2.32	0.214
5000	1.64	0.084	1.15	0.034	0.955	0.022	0.731	0.011					4.47	1.04	2.58	0.262
6000	1.96	0.118	1.38	0.049	1.15	0.031	0.877	0.016	18"				5.36	1.47	3.10	0.373
7000	2.29	0.158	1.61	0.065	1.34	0.042	1.02	0.021	0.808	0.012			6.25	2.0	3.61	0.499
8000	2.62	0.204	1.84	0.085	1.53	0.054	1.17	0.027	0.924	0.015			7.15	2.59	4.13	0.650
9000	2.95	0.256	2.08	0.107	1.72	0.067	1.31	0.033	1.04	0.019					4.65	0.816
10 000	3.28	0.313	2.31	0.130	1.91	0.081	1.46	0.041	1.15	0.023	20"				5.16	0.992
12 000	3.93	0.447	2.77	0.184	2.29	0.114	1.75	0.057	1.38	0.032	1.11	0.019	24"		6.20	1.41
14 000	4.59	0.600	3.23	0.246	2.67	0.153	2.05	0.077	1.62	0.044	1.30	0.025			7.23	1.91
16 000	5.24	0.776	3.69	0.317	3.06	0.198	2.34	0.099	1.85	0.056	1.49	0.032	1.03	0.013	8.26	2.48
18 000	5.90	0.975	4.15	0.398	3.44	0.246	2.63	0.124	2.08	0.069	1.67	0.040	1.16	0.016		
20 000	6.55	1.19	4.61	0.487	3.82	0.302	2.92	0.152	2.31	0.084	1.86	0.049	1.28	0.020		
25 000	8.19	1.83	5.77	0.758	4.77	0.469	3.65	0.234	2.89	0.130	2.32	0.076	1.61	0.030		
30 000			6.92	1.08	5.73	0.669	4.38	0.332	3.46	0.183	2.79	0.108	1.93	0.043		
35 000			8.07	1.46	6.68	0.903	5.12	0.446	4.04	0.248	3.25	0.144	2.25	0.057		
40 000			9.23	1.90	7.64	1.17	5.85	0.578	4.62	0.319	3.72	0.186	2.57	0.074		
45 000			10.38	2.39	8.59	1.47	6.58	0.726	5.19	0.400	4.18	0.233	2.89	0.092		
50 000					9.55	1.81	7.31	0.888	5.77	0.491	4.64	0.284	3.21	0.113		
55 000							8.04	1.07	6.35	0.594	5.11	0.343	3.53	0.136		
60 000							8.77	1.27	6.93	0.708	5.58	0.411	3.86	0.161		
65 000							9.5	1.49	7.50	0.822	6.04	0.475	4.18	0.189		
70 000							10.2	1.70	8.08	0.955	6.51	0.552	4.50	0.216		
75 000							11.0	1.98	8.66	1.10	6.97	0.628	4.82	0.246		

1 cubic meter = 1000 liters.

For pressure drop and velocity for pipe other than Schedule 40 and other than 100 meters long, see explanations on page B-15.

Flow of Water through Schedule 40 Steel Pipe,

continued from page B – 13

Pressure Drop for lengths of pipe other than 100 meters

For lengths of pipe other than 100 meters the pressure drop is proportional to the length. Thus, for 50 meters of pipe, the pressure drop is approximately one-half the value given in the table . . . for 300 meters, three times the given value, etc.

Velocity

Velocity is a function of the cross sectional flow area; thus, it is constant for a given flow rate and is independent of pipe length.

Pressure Drop and Velocity for pipe other than Schedule 40

To determine the velocity or pressure drop of water through pipe other than Schedule 40, use the following formulas:

$$v_a = v_{40}\left(\frac{d_{40}}{d_a}\right)^2$$

$$\Delta p_a = \Delta p_{40}\left(\frac{d_{40}}{d_a}\right)^5$$

Subscript "a" refers to the Schedule of pipe through which velocity or pressure drop is desired.

Subscript "40" refers to the velocity or pressure drop through Schedule 40 pipe, as given in the tables on page B-13.

Flow of Air through Schedule 40 Steel Pipe,

continued from facing page

Pressure Drop for lengths of pipe other than 100 meters

For lengths of pipe other than 100 meters the pressure drop is proportional to the length. Thus, for 50 meters of pipe, the pressure drop is approximately one-half the value given in the table . . . for 300 meters, three times the given value, etc.

The pressure drop is also inversely proportional to the absolute pressure and directly proportional to the absolute temperature.

Therefore, to determine the pressure drop for inlet or average pressures other than 7 bar and at temperatures other than 15 C, multiply the values given in the table by the ratio:

$$\left(\frac{7 + 1.013}{p + 1.013}\right)\left(\frac{273 + t}{288}\right)$$

where:

"p" is the inlet or average gauge pressure in bars, and,

"t" is the temperature in degrees Celsius under consideration.

Pressure Drop through pipe other than Schedule 40

To determine the pressure drop through pipe other than Schedule 40, use the following formula:

$$\Delta p_a = \Delta p_{40}\left(\frac{d_{40}}{d_a}\right)^5$$

Subscript "a" refers to the Schedule of pipe through which pressure drop is desired.

Subscript "40" refers to the pressure drop through Schedule 40 pipe, as given in the table on facing page.

Flow Rate of compressed air at temperature and pressure other than Metric Standard Conditions (MSC)

The cubic metres per minute of compressed air at any pressure is inversely proportional to the absolute pressure and directly proportional to the absolute temperature.

To determine the cubic metres per minute of compressed air at any temperature and pressure other than standard conditions (MSC), multiply the value of cubic metres per minute of free air by the ratio:

$$\left(\frac{1.013}{1.013 + p}\right)\left(\frac{2.73 + t}{288}\right)$$

Commercial Steel Pipe

Based on ANSI B36.10:1970 and BS 1600: Part 2: 1970

Schedule Wall Thicknesses

	Nominal Pipe Size Inches	Outside Diameter mm	Thickness mm	Inside Diameter mm
Schedule 10	14	355.6	6.35	342.9
	16	406.4	6.35	393.7
	18	457.2	6.35	444.5
	20	508.0	6.35	495.3
	24	609.6	6.35	596.9
	30	762.0	7.92	746.2
Schedule 20	8	219.1	6.35	206.4
	10	273.0	6.35	260.3
	12	323.9	6.35	311.2
	14	355.6	7.92	339.8
	16	406.4	7.92	390.6
	18	457.2	7.92	441.4
	20	508.0	9.52	489.0
	24	609.6	9.52	590.6
	30	762.0	12.70	736.6
Schedule 30	8	219.1	7.04	205.0
	10	273.0	7.80	257.4
	12	323.9	8.38	307.1
	14	355.6	9.52	336.6
	16	406.4	9.52	387.4
	18	457.2	11.13	434.9
	20	508.0	12.70	482.6
	24	609.6	14.27	581.1
	30	762.0	15.88	730.2
Schedule 40	1/8	10.3	1.73	6.8
	1/4	13.7	2.24	9.2
	3/8	17.1	2.31	12.5
	1/2	21.3	2.77	15.8
	3/4	26.7	2.87	21.0
	1	33.4	3.38	26.6
	1¼	42.2	3.56	35.1
	1½	48.3	3.68	40.9
	2	60.3	3.91	52.5
	2½	73.0	5.16	62.7
	3	88.9	5.49	77.9
	3½	101.6	5.74	90.1
	4	114.3	6.02	102.3
	5	141.3	6.55	128.2
	6	168.3	7.11	154.1
	8	219.1	8.18	202.7
	10	273.0	9.27	254.5
	12	323.9	10.31	303.3
	14	355.6	11.13	333.3
	16	406.4	12.70	381.0
	18	457.2	14.27	428.7
	20	508.0	15.09	477.8
	24	609.6	17.48	574.6
Schedule 60	8	219.1	10.31	198.5
	10	273.0	12.70	247.6
	12	323.9	14.27	295.4
	14	355.6	15.09	325.4
	16	406.4	16.64	373.1
	18	457.2	19.05	419.1
	20	508.0	20.62	466.8
	24	609.6	24.61	560.4
Schedule 80	1/8	10.3	2.41	5.5
	1/4	13.7	3.02	7.7
	3/8	17.1	3.20	10.7
	1/2	21.3	3.73	13.8
	3/4	26.7	3.91	18.9
	1	33.4	4.55	24.3
	1¼	42.2	4.85	32.5
	1½	48.3	5.08	38.1
	2	60.3	5.54	49.2
	2½	73.0	7.01	59.0
	3	88.9	7.62	73.7
Schedule 80—cont.	3½	101.6	8.08	85.4
	4	114.3	8.56	97.2
	5	141.3	9.52	122.3
	6	168.3	10.97	146.4
	8	219.1	12.70	193.7
	10	273.0	15.09	242.8
	12	323.9	17.47	289.0
	14	355.6	19.05	317.5
	16	406.4	21.44	363.5
	18	457.2	23.82	409.6
	20	508.0	26.19	455.6
	24	609.6	30.96	547.7
Schedule 100	8	219.1	15.09	188.9
	10	273.0	18.26	236.5
	12	323.9	21.44	281.0
	14	355.6	23.82	308.0
	16	406.4	26.19	354.0
	18	457.2	29.36	398.5
	20	508.0	32.54	442.9
	24	609.6	38.89	531.8
Schedule 120	4	114.3	11.13	92.0
	5	141.3	12.70	115.9
	6	168.3	14.27	139.8
	8	219.1	18.26	182.6
	10	273.0	21.44	230.1
	12	323.9	25.40	273.1
	14	355.6	27.79	300.0
	16	406.4	30.96	344.5
	18	457.2	34.92	387.4
	20	508.0	38.10	431.8
	24	609.6	46.02	517.6
Schedule 140	8	219.1	20.62	177.9
	10	273.0	25.40	222.2
	12	323.9	28.58	266.7
	14	355.6	31.75	292.1
	16	406.4	36.52	333.4
	18	457.2	39.69	377.8
	20	508.0	44.45	419.1
	24	609.6	52.39	504.8
Schedule 160	1/2	21.3	4.78	11.7
	3/4	26.7	5.56	15.6
	1	33.4	6.35	20.7
	1¼	42.2	6.35	29.5
	1½	48.3	7.14	34.0
	2	60.3	8.74	42.8
	2½	73.0	9.52	54.0
	3	88.9	11.13	66.6
	4	114.3	13.49	87.3
	5	141.3	15.88	109.5
	6	168.3	18.26	131.8
	8	219.1	23.01	173.1
	10	273.0	28.58	215.8
	12	323.9	33.34	257.2
	14	355.6	35.71	284.2
	16	406.4	40.49	325.4
	18	457.2	45.24	366.7
	20	508.0	50.01	408.0
	24	609.6	59.54	490.5

Commercial Steel Pipe

Based on ANSI B36.10:1970 and BS 1600 : Part 2 : 1970

Standard Wall Pipe

Nominal Pipe Size Inches	Outside Diameter mm	Thickness mm	Inside Diameter mm
1/8	10.3	1.73	6.8
1/4	13.7	2.24	9.2
3/8	17.1	2.31	12.5
½	21.3	2.77	15.8
¾	26.7	2.87	21.0
1	33.4	3.38	26.6
1¼	42.2	3.56	35.1
1½	48.3	3.68	40.9
2	60.3	3.91	52.5
2½	73.0	5.16	62.7
3	88.9	5.49	77.9
3½	101.6	5.74	90.1
4	114.3	6.02	102.3
5	141.3	6.55	128.2
6	168.3	7.11	154.1
8	219.1 S	8.18	202.7
10	273.0 S	9.27	254.5
12	323.9 S	9.52	304.9

Extra Strong Pipe

Nominal Pipe Size Inches	Outside Diameter mm	Thickness mm	Inside Diameter mm
1/8	10.3	2.41	5.5
1/4	13.7	3.02	7.7
3/8	17.1	3.20	10.7
½	21.3	3.73	13.8
¾	26.7	3.91	18.9
1	33.4	4.55	24.3
1¼	42.2	4.85	32.5
1½	48.3	5.08	38.1
2	60.3	5.54	49.2
2½	73.0	7.01	59.0
3	88.9	7.62	73.7
3½	101.6	8.08	85.4
4	114.3	8.56	97.2
5	141.3	9.52	122.3
6	168.3	10.97	146.4
8	219.1	12.70	193.7
10	273.0	12.70	247.6
12	323.9	12.70	298.5

Double Extra Strong Pipe

Nominal Pipe Size Inches	Outside Diameter mm	Thickness mm	Inside Diameter mm
½	21.3	7.47	6.4
¾	26.7	7.82	11.1
1	33.4	9.09	15.2
1¼	42.2	9.70	22.8
1½	48.3	10.16	28.0
2	60.3	11.07	38.2
2½	73.0	14.02	45.0
3	88.9	15.24	58.4
4	114.3	17.12	80.1
5	141.3	19.05	103.2
6	168.3	21.95	124.4
8	219.1	22.22	174.7
10	273.0	25.40	222.2
12	323.9	25.40	273.1

Stainless Steel Pipe

Based on ANSI B36.19-1965 and BS 1600 : Part 2 : 1970

Schedule 5 S

Nominal Pipe Size Inches	Outside Diameter mm	Thickness mm	Inside Diameter mm
½	21.3	1.65	18.0
¾	26.7	1.65	23.4
1	33.4	1.65	30.1
1¼	42.2	1.65	38.9
1½	48.3	1.65	45.0
2	60.3	1.65	57.0
2½	73.0	2.11	68.8
3	88.9	2.11	84.7
3½	101.6	2.11	97.4
4	114.3	2.11	110.1
5	141.3	2.77	135.8
6	168.3	2.77	162.8
8	219.1	2.77	213.6
10	273.0	3.40	266.2
12	323.9	3.96	316.0

Schedule 10 S

Nominal Pipe Size Inches	Outside Diameter mm	Thickness mm	Inside Diameter mm
1/8	10.3	1.24	7.8
¼	13.7	1.65	10.4
3/8	17.1	1.65	13.8
½	21.3	2.11	17.1
¾	26.7	2.11	22.5
1	33.4	2.77	27.9
1¼	42.2	2.77	36.7
1½	48.3	2.77	42.8
2	60.3	2.77	54.8
2½	73.0	3.05	66.9
3	88.9	3.05	82.8
3½	101.6	3.05	95.5
4	114.3	3.05	108.2
5	141.3	3.40	134.5
6	168.3	3.40	161.5
8	219.1	3.76	211.6
10	273.0	4.19	264.6
12	323.9	4.57	314.8

Schedule 40 S

1/8 to 12	Values are the same, size for size, as those shown on page B – 17 for Standard Wall Pipe

Schedule 80 S

1/8 to 12	Values are the same, size for size, as those shown on page B – 17 for Extra Strong Pipe.

Commercial Steel Pipe

Selected from ISO 336 - 1974 and BS 3600 : 1973

Nominal Pipe Size	Outside Diameter	Thickness	Inside Diameter	Nominal Pipe Size	Outside Diameter	Thickness	Inside Diameter	Nominal Pipe Size	Outside Diameter	Thickness	Inside Diameter
Inches	mm	mm	mm	Inches	mm	mm	mm	Inches	mm	mm	mm
1/8	10.2	1.6	7.0	1¼	42.4	3.2	36.0	3	88.9	5.4	78.1
		1.8	6.6			3.6	35.2			5.6	77.7
		2.0	6.2			4.0	34.4			5.9	77.1
		2.3	5.6			4.5	33.4			6.3	76.3
						5.0	32.4			7.1	74.7
¼	13.5	1.8	9.9			5.4	31.6			8.0	72.9
		2.0	9.5			5.6	31.2			8.8	71.3
		2.3	8.9			5.9	30.6			10.0	68.9
		2.6	8.3			6.3	29.8			11.0	66.9
		2.9	7.7			7.1	28.2			12.5	63.9
						8.0	26.4			14.2	60.5
						8.8	24.8			16.0	56.9
3/8	17.2	2.0	13.2			10.0	22.4*				
		2.3	12.6								
		2.6	12.0	1½	48.3	3.2	41.9	3½	101.6	5.6	90.4
		2.9	11.4			3.6	41.1			5.9	89.8
		3.2	10.8			4.0	40.3			6.3	89.0
						4.5	39.3			7.1	87.4
						5.0	38.3			8.0	85.6
½	21.3	2.6	16.1			5.4	37.5			8.8	84.0
		2.9	15.5			5.6	37.1			10.0	81.6
		3.2	14.9			5.9	36.5			11.0	79.6
		3.6	14.1			6.3	35.7			12.5	76.6
		4.0	13.3			7.1	34.1			14.2	73.2
		4.5	12.3			8.0	32.3			16.0	69.6
		5.0	11.3			8.8	30.7			17.5	66.6
		5.4	10.5			10.0	28.3				
								4	114.3	5.6	103.1
¾	26.9	2.6	21.7	2	60.3	3.6	53.1			5.9	102.5
		2.9	21.1			4.0	52.3			6.3	101.7
		3.2	20.5			4.5	51.0			7.1	100.1
		3.6	19.7			5.0	50.3			8.0	98.3
		4.0	18.9			5.4	49.5			8.8	96.7
		4.5	17.9			5.6	49.1			10.0	94.3
		5.0	16.9			5.9	48.5			11.0	92.3
		5.4	16.1			6.3	47.7			12.5	89.3
		5.6	15.7			7.1	46.1			14.2	85.9
		5.9	15.1			8.0	44.3			16.0	82.3
		6.3	14.3			8.8	42.7			17.5	79.3
		7.1	12.7			10.0	40.3			20.0	74.3
						11.0	38.3				
1	33.7	3.2	27.3	2½	76.1	5.0	66.1	5	139.7	5.9	127.9
		3.6	26.5			5.4	65.3			6.3	127.1
		4.0	25.7			5.6	64.9			7.1	125.5
		4.5	24.7			5.9	64.3			8.0	123.7
		5.0	23.7			6.3	63.5			8.8	122.1
		5.4	22.9			7.1	61.9			10.0	119.7
		5.6	22.5			8.0	60.1			11.0	117.7
		5.9	21.9			8.8	58.5			12.5	114.7
		6.3	21.1			10.0	56.1			14.2	111.3
		7.1	19.5			11.0	54.1			16.0	107.7
		8.0	17.7			12.5	51.1			17.5	104.7
		8.8	16.1			14.2	47.7			20.0	99.7

* Not included in BS 3600 : 1973

Commercial Steel Pipe – continued

Nominal Pipe Size	Outside Diameter	Thickness	Inside Diameter	Nominal Pipe Size	Outside Diameter	Thickness	Inside Diameter
Inches	mm	mm	mm	Inches	mm	mm	mm
6	168.3	6.3	155.7	16	406.4	6.3	393.8
		7.1	154.1			7.1	392.2
		8.0	152.3			8.0	390.4
		8.8	150.7			8.8	388.8
		10.0	148.3			10.0	386.4
		11.0	146.3			11.0	384.4
		12.5	143.3			12.5	381.4
		14.2	139.9			14.2	378.0
		16.0	136.3			16.0	374.4
		17.5	133.3			17.5	371.4
		20.0	128.3			20.0	366.4
		22.2	123.9			22.2	362.0
						25.0	356.4
8	219.1	6.3	206.5			28.0	350.4*
		7.1	204.9			30.0	346.4*
		8.0	203.1			32.0	342.4*
		8.8	201.5			36.0	334.4*
		10.0	199.1			40.0	326.4*
		11.0	197.1			45.0	316.4*
		12.5	194.1				
		14.2	190.7	18	457.0	6.3	444.4
		16.0	187.1			7.1	442.8
		17.5	184.1			8.0	441.0
		20.0	179.1			8.8	439.4
		22.2	174.7			10.0	437.0
		25.0	169.1			11.0	435.0
						12.5	432.0
10	273.0	6.3	260.4			14.2	428.6
		7.1	258.8			16.0	425.0
		8.0	257.0			17.5	422.0
		8.8	255.4			20.0	417.0
		10.0	253.0			22.2	412.6
		11.0	251.0			25.0	407.0
		12.5	248.0			28.0	401.0*
		14.2	244.6			30.0	397.0*
		16.0	241.0			32.0	393.0*
		17.5	238.0			36.0	385.0*
		20.0	233.0			40.0	377.0*
		22.2	228.6			45.0	367.0*
		25.0	223.0			50.0	357.0*
		28.0	217.0*				
		30.0	213.0*	20	508.0	6.3	495.4
						7.1	493.8
12	323.9	6.3	311.3			8.0	492.0
		7.1	309.7			8.8	490.4
		8.0	307.9			10.0	488.0
		8.8	306.3			11.0	486.0
		10.0	303.9			12.5	483.0
		11.0	301.9			14.2	479.6
		12.5	298.9			16.0	476.0
		14.2	295.5			17.5	473.0
		16.0	291.9			20.0	468.0
		17.5	288.9			22.2	463.6
		20.0	283.9			25.0	458.0
		22.2	279.5			28.0	452.0*
		25.0	273.9			30.0	448.0*
		28.0	267.9*			32.0	444.0*
		30.0	263.9*			36.0	436.0*
		32.0	259.9*			40.0	428.0*
		36.0	251.9*			45.0	418.0*
						50.0	408.0*
14	355.6	6.3	343.0			55.0	398.0*
		7.1	341.4				
		8.0	339.6	24	610.0	6.3	597.4
		8.8	338.0			7.1	595.8
		10.0	335.6			8.0	594.0
		11.0	333.6			8.8	592.4
		12.5	330.6			10.0	590.0
		14.2	327.2			11.0	588.0
		16.0	323.6			12.5	585.0
		17.5	320.6			14.2	581.6
		20.0	315.6			16.0	578.0
		22.2	311.2			17.5	575.0
		25.0	305.6			20.0	570.0
		28.0	299.6*			22.2	565.6
		30.0	295.6*			25.0	560.0
		32.0	291.6*			28.0	554.0*
		36.0	283.6*			30.0	550.0*
						32.0	546.0*
						36.0	538.0*
						40.0	530.0*
						45.0	520.0*
						50.0	510.0*

* Not included in BS 3600 : 1973